

# traces-sm — SGX Secrets & Key Management Console

Eight screens of the cross-platform native desktop client (eframe/egui on GTK3, WebView2 and WKWebView), captured from the interactive design. Navigation is a grouped sidebar with breadcrumb history, a ⌘K command palette and a local Traces AI assistant panel.

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traces-sm / Overview / Dashboard

Q Search keys, secrets, nodes...

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ENCLAVE DEVICE

/dev/sgx\_enclave

HW\_ACTIVE

EPC 64 MB

eframe/egui

Ubuntu 24.04 (GTK3)

RA-TLS VERIFIED

EPC 21.2 MB / 64.0 MB

ACTIVE NIST KEYS

8

RSA · ECDSA · ML-KEM

SEALED SECRETS

14

AES-256-GCM envelope

DKG PEER QUORUM

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2-of-3 · RA-TLS connected

ENCLAVE EPC MEMORY & CRYPTOPERIOD TELEMETRY

EPC memory

21.2 MB / 64.0 MB

Cryptoperiod volume

1.2 GB / 4.2 GB (28.5%)

Hardware sealing key

EGETKEY · MRSIGNER bound

CRYPTOGRAPHIC THROUGHPUT — OPS/SEC

peak 937

00:00

06:00

12:00

18:00

ARCHITECTURE STACK

Enclave

Fortanix EDP · Rust SGX

Host

Axum 0.7 · Tokio · SQLite

Desktop

eframe / egui · GTK3 · WebView2 · WKWebView

Web GUI

Yew 0.21 · wasm32

CLI

clap 4.5 · comfy-table

Traces AI

Local model · no secrets leave the host

×

Enclave is HW\_ACTIVE and all six policy invariants are enforcing. Ask me about a key, a policy decision, or an attestation quote.

Why k-104?

APT cutoff

Quorum health

Ask about keys, policy, attestation...

Send

traces-sm — SGX HW\_ACTIVE

RA-TLS VERIFIED

Traces AI

admin-role · signed in

**Contents** — four summary cards (active NIST keys, sealed secrets, DKG quorum, DRBG health), the EPC memory and cryptoperiod meters, throughput chart and architecture stack.

**Live** — EPC gauge and peak ops/sec update from enclave telemetry.

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traces-sm — SGX Secrets & Key Management

Q Search keys, secrets, nodes...

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traces-sm / Keys & secrets / Key Lifecycle

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ENCLAVE DEVICE

/dev/sgx\_enclave

HW\_ACTIVE

EPC 64 MB

eframe/egui · Ubuntu 24.04 (GTK3) RA-TLS VERIFIED EPC 19.0 MB / 64.0 MB

Key lifecycle matrix

NIST SP 800-57 Part 1 Rev. 5 — pre-operational through destroyed.

All

Pre-Operational

Operational

Deactivated

Revoked

Shredded

| KEY ALIAS & ID         | ALGORITHM  | KEYUSAGE      | SIGN   VERIFY   | CRYPTOPERIOD    | NIST STATE      | ACTION   |
|------------------------|------------|---------------|-----------------|-----------------|-----------------|----------|
| rsa-master-cert k-101  | RSA-4096   | SIGN   VERIFY | 1.2 GB / 4.2 GB | 1.2 GB / 4.2 GB | OPERATIONAL     | Rotate   |
| ecdsa-token-sign k-102 | ECDSA-P256 | SIGN   VERIFY | 450 MB / 4.2 GB | 450 MB / 4.2 GB | OPERATIONAL     | Rotate   |
| pqc-kyber-kem k-103    | ML-KEM-768 | KEY_WRAP      | 0 B / 4.2 GB    | 0 B / 4.2 GB    | PRE_OPERATIONAL | Activate |
| legacy-rsa-2048 k-104  | RSA-2048   | DECRYPT_ONLY  | 4.2 GB / 4.2 GB | 4.2 GB / 4.2 GB | DEACTIVATED     | Shred    |

SELECTED KEY — LEGACY-RSA-2048 (K-104)

State

DEACTIVATED

Reason

Cryptoperiod volume limit reached (2^32 bytes).

Policy restriction

Operational signing disabled. Historical decryption permitted.

Export public PEM

View attestation binding

+ Generate key

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Send

admin-role · signed in

**Contents** — state filters across the Part 1 Rev. 5 machine, the key table with KeyUsage flags and cryptoperiod meters, and the selected-key detail strip.

**Actions** — activate, rotate, SP 800-88 crypto-shred, export public PEM.

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Keys & secrets

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ENCLAVE DEVICE

/dev/sgx\_enclave

HW\_ACTIVE

EPC 64 MB

eframe/egui · Ubuntu 24.04 (GTK3)    RA-TLS VERIFIED    EPC 19.0 MB / 64.0 MB    admin-role · signed in

traces-sm

Ubuntu 24.04 (GTK3)

SGX HW\_ACTIVE

RA-TLS VERIFIED

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Traces AI

Local model · no secrets leave the host

Enclave is HW\_ACTIVE and all six policy invariants are enforcing. Ask me about a key, a policy decision, or an attestation quote.

+ Create

Search by name, tag, or owner...

All

Opaque

SymmetricKey

CertBundle

| SECRET NAME                | TYPE         | VER | OWNER      | EXPIRATION | SEALING     | REVEAL |
|----------------------------|--------------|-----|------------|------------|-------------|--------|
| db-master-pass<br>sec-901  | Opaque       | v2  | admin-role | 2027-12-31 | AES-256-GCM | Reveal |
| aws-access-key<br>sec-902  | SymmetricKey | v1  | deploy-bot | 2026-11-15 | AES-256-GCM | Reveal |
| prod-tls-bundle<br>sec-903 | CertBundle   | v3  | sec-ops    | 2026-09-01 | AES-256-GCM | Reveal |

Reveal decrypts inside the SGX enclave against an active authorization token; plaintext is zeroized when the drawer closes.

Why k-104?

APT cutoff

Quorum health

Ask about keys, policy, attestation...

Send

**Contents** — search and type filter, the sealed-secret table with version, owner, expiry and AES-256-GCM sealing badge, and per-row reveal.

**Modal** — create-secret dialog with payload and TTL, sealed in-enclave.

FileEnclaveKeysVaultToolsHelp

traces-sm / CryptographyEntropy

Q Search keys, secrets, nodes...⌕

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traces-sm — SGX Secrets & Key Management

RA-TLS VERIFIED

SGX HW\_ACTIVE

Ubuntu 24.04 (GTK3)

Traces AI

admin-role

ENCLAVE DEVICE /dev/sgx\_enclaveHW\_ACTIVEEPC 64 MB

eframe/egui · Ubuntu 24.04 (GTK3)RA-TLS VERIFIEDEPC 19.1 MB / 64.0 MB

DRBG & entropy health

NIST SP 800-90A/B/C continuous health testing on the hardware entropy source.

Source: SGX RDRAND / ROSEEDDRBG: HMAC\_DRBG (SHA-256)

Re-seed DRBG

PASS

Repetition Count Test (RCT)

StandardSP 800-90B §4.4.1Continuous

Sample window16

Cutoff threshold C3

Current repetition count

No drift detected.

PASS

Adaptive Proportion Test (APT)

StandardSP 800-90B §4.4.2512 samples

Window size W13

Cutoff threshold C5

Max sample frequency

Uniform entropy across the window.

RESEED COUNTERS & DRBG AUDIT LOG

Total reseed iterations1,048,576

2026-08-25 10:52:14 UTCDRBG reseeded with 32-byte SGX ROSEED hardware entropy vector.

2026-08-25 09:41:02 UTCContinuous RCT/APT health tests completed on 512-sample window.

2026-08-25 08:15:44 UTCEntropy source re-qualified against SP 800-90B §4.4 cutoffs.

Last reseed timestamp2026-08-25 10:52:14 UTC

Enclave is HW\_ACTIVE and all six policy invariants are enforcing. Ask me about a key, a policy decision, or an attestation quote.

Why k-104?APT cutoffQuorum health

Ask about keys, policy, attestation...

Send

Contents — threshold selector (2-of-3, 3-of-5, 5-of-7), the RA-TLS peer topology map with DCAP attestation badges, and protocol round telemetry.

Protocol — Pedersen VSS / FROST share exchange.

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ENCLAVE DEVICE

/dev/sgx\_enclave

HW\_ACTIVE

EPC 64 MB

eframe/egui · Ubuntu 24.04 (GTK3) RA-TLS VERIFIED EPC 20.0 MB / 64.0 MB

UBUNTU 24.04 (GTK3)

SGX HW\_ACTIVE

RA-TLS VERIFIED

Traces AI

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Local model · no secrets leave the host

Enclave is HW\_ACTIVE and all six policy invariants are enforcing. Ask me about a key, a policy decision, or an attestation quote.

Mandatory security policy

Governance invariants enforced by enclave/sic/policy.is.

Active profile

StrictNistProfile

CommercialCloudProfile

CustomPolicy

SP 800-57 / FIPS 140-3

Save policy

IN-MEMORY & IN-STORAGE POLICY INVARIANTS

☒

FIPS 140-3 RAM zeroization

zeroize::Zeroizing<T> volatile scrub

☒

Enclave EPC hardware page encryption

Intel SGX memory controller

☒

NIST SP 800-38F key wrapping

prohibit plaintext private key export

☒

NIST SP 800-57 cryptoperiod byte limits

2^32 bytes max per key

☒

NIST SP 800-88 crypto-shredding

sector overwrite before delete

☒

Remote attestation quotes

Intel DCAP verification required for P2P mesh

ENFORCING 6 of 6 security invariants active in the SGX enclave.

Why k-104?

APT cutoff

Quorum health

Ask about keys, policy, attestation...

Send

admin-role · signed in

Contents — the SP 800-90B RCT and APT test cards with live counts against their cutoffs, reseed counters and the DRBG audit log.

Action — manual re-seed from RDSEED hardware entropy.

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RSA · ECDSA · ML-KEM

SEALED SECRETS

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AES-256-GCM envelope

DKG PEER QUORUM

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2-of-3 · RA-TLS connected

SP 800-90B DRBG

HEALTHY

APT & RCT passed

ENCLAVE EPC MEMORY & CRYPTOPERIOD TELEMETRY

EPC memory

19.1 MB / 64.0 MB

Cryptoperiod volume

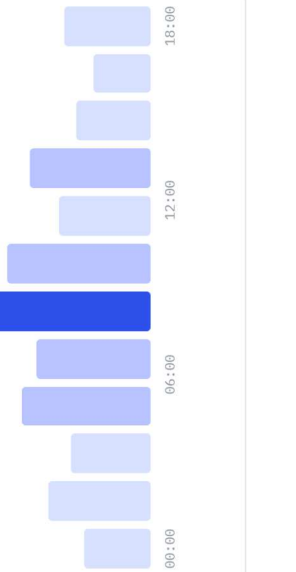
1.2 GB / 4.2 GB (28.5%)

EGETKEY · MRSIGNER bound

Hardware sealing key

CRYPTOGRAPHIC THROUGHPUT — OPS/SEC

peak 947



ENCLAVE DEVICE

/dev/sgx\_enclave

HW\_ACTIVE

EPC 64 MB

eframe/egui · Ubuntu 24.04 (GTK3)

RA-TLS VERIFIED

EPC 19.1 MB / 64.0 MB

Traces AI

Local model · no secrets leave the host

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Why k-104?

APT cutoff

Quorum health

Ask about keys, policy, attestation...

Send

Contents — sub-tabs for Schnorr PoK, Bulletproofs range proofs and Paillier PHE, the proof generator with verification result, and the homomorphic addition demo.

Execution — all proofs generated inside the enclave.



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ENCLAVE DEVICE /dev/sgx\_enclave

HW\_ACTIVE

EPC 64 MB

ENCLAVE HEALTH

Enclave health, key counts, cryptoperiod limits and cryptographic throughput.

ACTIVE NIST KEYS 8

SEALD SECRETS 14

DKG PEER QUORUM 3

SP 800-90B DRBG HEALTHY

ENCLAVE EPC MEMORY & CRYPTOPERIOD TELEMETRY

EPC memory 17.5 MB / 64.0 MB

Cryptoperiod volume 1.2 GB / 4.2 GB (28.5%)

Hardware sealing key EGETKEY · MRSIGNER bound

CRYPTOGRAPHIC THROUGHPUT — OPS/SEC

peak 922

00:00 06:00 12:00 18:00

ARCHITECTURE STACK

Enclave Fortanix EDP · Rust SGX

Host Axum 0.7 · Tokio · SQLite

Desktop eframe / egui · GTK3 · WebView2 · WkWebView

Web GUI Yew 0.21 · wasm32

CLI clap 4.5 · comfy-table

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eframe/egui · Ubuntu 24.04 (GTK3)

RA-TLS VERIFIED

EPC 17.5 MB / 64.0 MB

admin-role · signed in

**Contents** — profile selector, the six policy invariants as toggles (zeroization, EPC encryption, key wrapping, cryptoperiod limits, crypto-shredding, attestation) and the enforcement badge.

**Source** — [enclave/src/policy.rs](#).



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Audit logs & attestation quotes

Non-repudiable audit trail and raw Intel DCAP quote inspection.

All Actions

CREATE\_KEY

ROTATE\_KEY

SHRED\_KEY

Filter logs...

Export audit CSV

Last 24 hours

| TIMESTAMP (UTC)     | PRINCIPAL  | ACTION        | RESOURCE | SOURCE IP    | POLICY RESULT      |
|---------------------|------------|---------------|----------|--------------|--------------------|
| 2026-08-25 10:51:04 | admin-role | CREATE_KEY    | k-101    | 192.168.1.10 | PERMIT (NIST)      |
| 2026-08-25 10:48:22 | deploy-bot | ROTATE_KEY    | k-102    | 192.168.1.15 | PERMIT (NIST)      |
| 2026-08-25 10:44:57 | sec-ops    | REVEAL_SECRET | sec-901  | 192.168.1.20 | PERMIT (RA-TLS)    |
| 2026-08-25 10:30:11 | sec-ops    | SHRED_KEY     | k-104    | 192.168.1.20 | PERMIT (SP800-88)  |
| 2026-08-25 10:12:03 | admin-role | RESEED_DRBG   | dtbg-0   | 192.168.1.10 | PERMIT (SP800-90B) |

INTEL DCAP REMOTE ATTESTATION QUOTE

MRENCLAVE

0x8a91f4b... (32 B)

MRSIGNER

1 Provider

0x41f89a... (32 B)

Intel PCCS

VERIFIED\_SUCCESSFUL

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/dev/sgx\_enclave

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eframe/egui · Ubuntu 24.04 (GTK3) RA-TLS VERIFIED EPC 19.7 MB / 64.0 MB

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